Giovanni **Ranieri**

Profile: Master Student at EPFL in **Communication Systems**. High interest in Mathematics applied to communications, space, embedded systems and wireless technologies.

Phone: +41 79 152 78 42
Email: giovanni.ranieri@epfl.ch
Portfolio: www.flxinxout.github.io

WORK EXPERIENCE

Vice-President Research - EPFL Xplore | 2025-2026

Lead the EPFL Xplore Research pole. Strategy, event programming, and external representation of the association.

Software System Enginner - EPFL Xplore | 2024-2025

Winner of the European Rover Challenge 2025. Responsible for the entire software system of a Martian-like Rover and Drone, leading a team of 20 bachelor and master students in an international competition in Poland.

Student Assistant - VILAB, EPFL | Fall 2023

Contributed to the creation of the new IC Bachelor course "*Communication System Project*"

Control Station Member - EPFL Xplore | 2023-2024

Control Station web developer and Mikrotik configurations with RouterOS.

EDUCATION

EPFL, Swiss Federal Institute of Technology Lausanne | 2024-PRESENT

Master of Science in Communication Systems, Wireless Communication Specialization

EPFL, Swiss Federal Institute of Technology Lausanne | 2020-2024

Bachelor of Science in Communication Systems - GPA **5**/6

Gymnase in Nyon | 2017-2020

High School (Baccalauréat) - Biology and Chemistry

PROJECTS

Joint Communication & Sensing on RFSoC Phased Arrays | Spring 2026

Deployed and experimentally evaluated a JCAS architecture on a ZCU208 RFSoC, reproducing communication and sensing experiments and extending the system with fine-grained RFSoC-driven phased-array phase control.

Evaluation of NPU Acceleration for Offloading 5G Software Workloads | Spring 2025

FFT Implementation on AMD/Xilinx XDNA/AIE-ML Neural Processing Unit using MLIR-AIE. Evaluation of performance compared to CPU implementation from srsRAN.

Speeding up the Quantification of Data Staleness in DBO | Spring 2024

C++ Implementation of a critetion for a well-suited optimizer for dynamic black-box functions. Evaluation of performance compared to state of the art methods.

- Contribution to NeurIPS'24 Paper "*This too shall pass: Removing Stale Observations in Dynamic Bayesian Optimization*"

Flipper – A Snake Robot | Spring 2023

A 1m20 long Snake Robot designed and built at EPFL (Making Intelligent Things course).

- *Invited to present at annual EPFL IC Section Meeting in Lausanne*
- *Invited to showcase at Viscon2023 in ETH*
- Interviewed for EPFL News

SKILLS

Programming & Technologies

- C/C++, Java, Python, JavaScript, Assembly, VHDL/Verilog
- Docker, Kubernetes, Redis, MongoDB, RabbitMQ, ReactJS, ROS2, AWS

Mathematics & Computer Science

- Statistical Signal Processing, Stochastic Models
- Digital Communications, Mobile Networks, Satellite Theory
- Algorithms, Computer Architecture

Management

- EPFL Xplore (Research & Competition Poles)

Languages

- French (Native Speaker)
- Italien (Second Language)
- English (Fluent)

INTERESTS

Mathematics & Communication Systems: DSP, Mobile Networks, Navigation

Programming & Tools: C/C++, ROS2, VHDL/Verilog, Python, Assembly, Docker

Leadership: Managed timelines and integration between software subsystems

Languages: French & Italien (Native), English (Fluent)

CERTIFICATES

European Rover Challenge | 2024 & 2025: Winner 2025 Edition as Lead Software

OTHER ACTIVITIES

Sports: Climbing, Swimming

Students 4 Students: Student Assistant for Analysis and Physics in a preparation week for new EPFL students